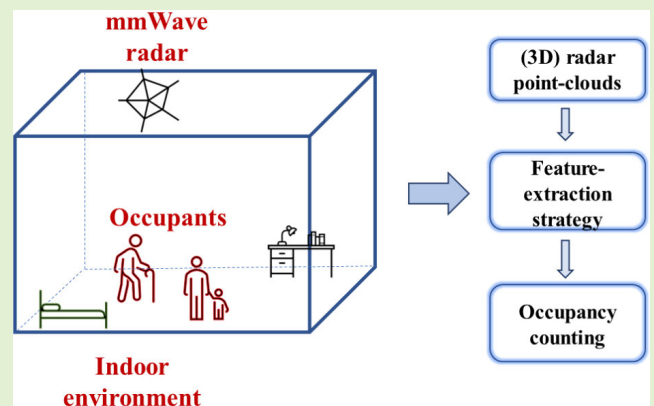


Novel Robust On-Line Indoor Occupancy Counting System Using mmWave Radar

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Abstract—In this work, a novel robust on-line indoor occupancy counting approach is proposed using the millimeter-wave (mmWave) frequency-modulated continuous-wave (FMCW) radar. The acquired radar data can be represented as sparse 3-D radar point clouds. The conventional indoor occupancy counting schemes suffer from various types of errors and uncertainties emerging from mmWave radar sensors. Thus, we propose a novel feature extraction strategy to obtain the robust features for training the machine-learning-driven occupancy counter. We formulate the underlying indoor occupancy counting problem as the multiclassification problem such that the k nearest-neighbor (KNN) classifier is adopted to identify the number of occupants on line. Experimental results from real-world data demonstrate that our proposed new approach leads to a promising classification accuracy of 95.8% for indoor occupancy counting. In the comparative study based on real-world data, our proposed novel indoor occupancy counting method greatly outperforms other existing schemes.

Index Terms—Indoor occupancy counting, k nearest-neighbor (KNN) classifier, machine learning, millimeter-wave (mmWave) frequency-modulated continuous-wave (FMCW) radar.



I. INTRODUCTION

THE *occupancy counting problem* is of paramount importance to many practical applications, such as remote patient monitoring and surveillance for smart health care [1], building control for safety and management [2], [3], and human object detection and rescue for emergency [4]. Videos, color images, and infrared (IR) images are usually acquired as the essential input features for automatic occupancy counting

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in [3], [5], [6], [7], [8], [9], [10], and [11]. However, video and image data are often sensitive especially due to privacy concern. On the other hand, wireless devices based on WiFi, Zigbee, radio frequency identification (RFID), and Bluetooth technologies have been broadly used in our daily life. Radio-activity fingerprints such as received signal-strength indicator (RSSI) and channel state information (CSI) produced by the aforementioned wireless devices have also been explored for automatic occupancy counting [12], [13], [14], [15], [16]. Nonetheless, both RSSI and CSI are very sensitive to the environment, and hence the learning systems based on these radio-activity fingerprints cannot be generalized when the floor plan is changed (e.g., addition/removal of furniture, a new room, or a new building). Whenever the environment (even slightly) changes, tremendous new training data must be acquired to accommodate this change, and therefore the aforementioned radio-activity-based schemes definitely need to be retrained for different environments, which are very impractical. Due to the difficulty in generalization from limited training, researchers have been seeking promising alternatives to the radio-activity fingerprint-based schemes.

Recently, radars have been emerging as the popular devices for tracking indoor moving objects, identifying gestures and postures of humans, and counting occupants [14], [17], [18].

In comparison to video cameras, the striking advantage of radars is that the privacy can be protected as radar point clouds cannot be used to identify individuals [19]. Furthermore, cameras may not be effective for certain scenarios due to their limitations such as blind spots [20] and lack of lighting [21]. On the other hand, radar sensors can be very robust especially against the aforementioned limitations of cameras [22]. Thus, radar sensors have become more and more popular for indoor occupancy counting recently [23], [24], [25]. The reflected signals from the human objects captured by radar sensors are adopted as the essential measures to estimate the number of occupants using various machine learning methods [26], [27], [28], [29]. For instance, the support vector machine (SVM) method was adopted as either a multiclassifier to determine the number of occupants using the high-resolution radar data in [27] or a regressor to predict the number of the moving objects in [26]. The decision tree (DT) (see [29]) and random forest (RF) (see [28]) algorithms were also applied to build the multiclassifiers for indoor occupancy counting based on radar data. A promising clustering scheme, namely, “density-based spatial clustering of applications with noise (DBSCAN),” was used to cluster the radar point clouds and estimate the number of indoor occupants [30], [31], [32]. Lately, deep neural networks (DNNs) were adopted to combat the indoor occupancy counting problem [14], [33], [34], [35], [36], [37] but they could not perform well due to the errors and uncertainties such as imprecise measurements caused by low signal-to-noise ratio (SNR) scenarios and the large distance between the radar and the target, which often arise in practice [38]. Besides, these deep learning methods require tremendous data and high computational complexity for training. Therefore, in this work, we would like to exploit a novel feature extraction mechanism for radar data and establish a new robust indoor occupancy counting approach, which does not rely on tremendous training data, needs not to be retrained for different environments, and can be generalized easily for different occupants’ geometric information.

II. RELATED WORKS AND OUR MAIN CONTRIBUTIONS

In this work, the TI-IWR6843 millimeter-wave (mmWave) frequency-modulated continuous-wave (FMCW) radar manufactured by Texas Instruments (TI) is used to acquire the radar point cloud data. Therefore, this section is mainly focused on the related works using the point cloud data acquired by the TI-IWR6843 mmWave FMCW radar for indoor occupancy counting. Here, we would like to highlight the current indoor occupancy counting challenges and manifest the main contributions of our proposed new approach in this article.

An existing indoor occupancy counting method, namely, “WIMOC,” uses the Kalman filter for the point cloud data acquired by the TI-IWR6843 mmWave FMCW radar to estimate the number of indoor occupants [31]. However, the “WIMOC” method fails to distinguish between two and three occupants. The “cTracker” method, which is based on the DBSCAN clustering technique, has been proposed for indoor occupancy counting [32]. Nevertheless, the “cTracker” method fails to distinguish two or three human objects walking fairly closely together. Furthermore, a TI built-in people-counting

function called “Gtrack” can be accessed through the TI-IWR6843 mmWave FMCW radar platform [39]. According to our empirical experience, the “Gtrack” technique leads to high accuracy when only one human object is present.

To combat the drawbacks of the existing methods, we propose a novel robust on-line indoor occupancy counting approach in this work. The primary contributions of our work can be summarized as follows.

- 1) We propose a novel robust on-line indoor occupancy counting scheme using point cloud data acquired by the TI-IWR6843 mmWave FMCW radar.
- 2) We propose a new feature extraction method to obtain robust input features for the k nearest-neighbor (KNN) classifier, which can be much less insensitive to various types of errors and uncertainties emerging from radar data.
- 3) Our proposed new indoor occupancy counting approach can correctly distinguish multiple occupants (such as two or three) which the existing methods fail to.

The rest of this article is organized as follows. The details of our proposed novel robust on-line indoor occupancy counting method are introduced in Section III, which consists of three major mechanisms, namely: 1) data acquisition; 2) data processing; and 3) KNN classifier. Our experimental results are presented and analyzed in Section IV. Meanwhile, the results from our proposed new scheme are also compared with those from the existing “WIMOC,” “cTracker,” and “Gtrack” methods in Section IV. Conclusion will be ultimately drawn in Section V.

III. OUR PROPOSED METHOD

Here, we introduce our novel robust on-line indoor occupancy counting system, which consists of three primary mechanisms: 1) data acquisition; 2) data processing; and 3) KNN classifier. A mmWave FMCW radar is first used to transmit the electromagnetic waves to hit human objects. By capturing the reflected signals, the mmWave radar system can determine the range(s) and angles associated with the human object(s). Such range(s) and angles can be further transformed into (sparse) 3-D point clouds. It is well-known that there often exist misleading outliers and noisy points in radar point clouds. Here, we propose a novel feature extraction technique to obtain robust input features for indoor occupancy counting. Then, such input features will be used to train a KNN classifier. The detailed steps are presented in Sections III-A–III-C.

A. Data Acquisition

We use the TI-IWR6843 mmWave FMCW radar to acquire the reflected signals from the human object(s). The mmWave FMCW radar is shown by Fig. 1. The system parameters of the adopted TI-IWR6843 mmWave FMCW radar are listed in Table I according to [39]. Furthermore, a Kinect V2 sensor is also facilitated to obtain the actual number of occupants as the ground truth. The experimental settings of data acquisition are stated as follows. The laboratory is a 11.8×7.5 m rectangular room, which contains three desks and a mmWave radar placed in the center of the ceiling. The height of the mmWave radar

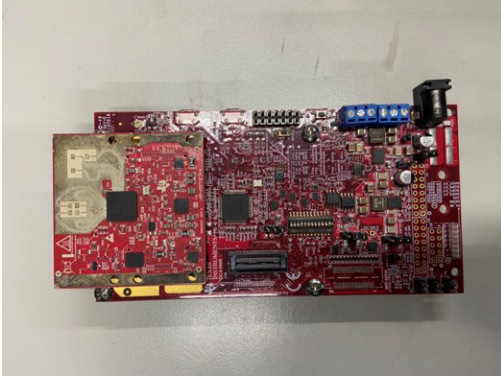


Fig. 1. Photograph of our adopted TI-IWR6843 mmWave FMCW radar.

TABLE I

SYSTEM PARAMETERS OF TI-IWR6843 mmWave FMCW RADAR

Parameters	Values
Starting Frequency (GHz)	60.75
Maximum Range (m)	8.1
Range Resolution (cm)	8.4
Maximum Velocity (km/h)	16.2
Velocity Resolution (km/h)	0.324
Ramp Slope (MHz/ μ s)	57
Number of Samples per Chirp	96
Number of Chirps	288
Idle Time (μ s)	30

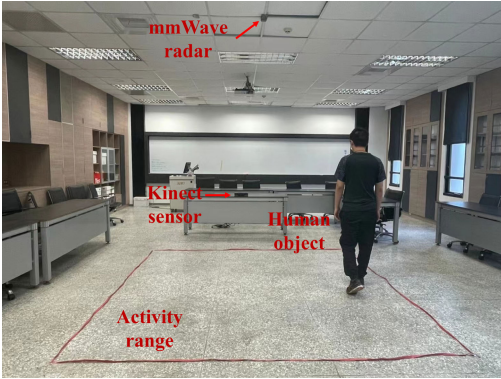


Fig. 2. Laboratory layout adopted for our experiment.

we have installed is 2.9 m. The Kinect V2 device is placed at the center of a desk. The human objects are moving within a 3×3 m square area in the middle of the laboratory. The floor plan (layout) of our laboratory is illustrated by Fig. 2.

For each frame (a snapshot in time), the range(s) and angles of the human object(s) can be obtained from the reflected signals by the mmWave radar. In practice, for each frame, say at time t , according to [39], the range(s) of the human object(s), R_t^d , and the angles of the human object(s), namely, the elevation angle(s) θ_t^d and the azimuth angle(s) ψ_t^d , can be converted into the 3-D radar point (x_t^d, y_t^d, z_t^d) such that

$$x_t^d \stackrel{\text{def}}{=} R_t^d \cos(\psi_t^d) \sin(\theta_t^d) \quad (1)$$

$$y_t^d \stackrel{\text{def}}{=} R_t^d \cos(\psi_t^d) \cos(\theta_t^d) \quad (2)$$

$$z_t^d \stackrel{\text{def}}{=} R_t^d \sin(\psi_t^d) \quad (3)$$

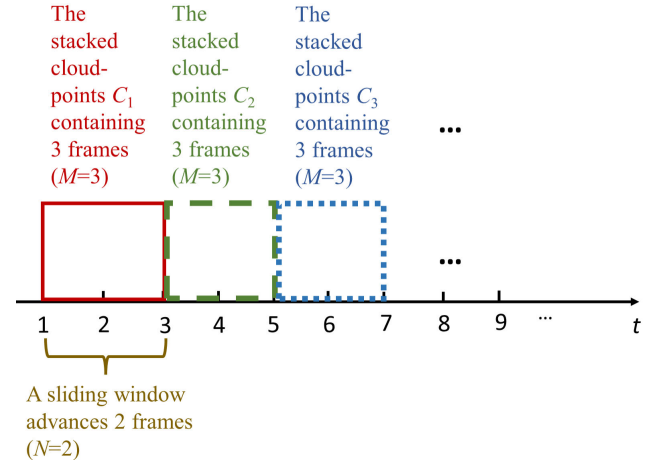


Fig. 3. Example of the stacking and sliding procedure for $M = 3$ and $N = 2$.

where $d = 0, 1, \dots, D_t - 1$, and D_t specifies the number of detected points in the 3-D radar point cloud at each time (frame) t . Note that the number of detected points D_t in the radar point cloud of a frame t varies with respect to the mobility of the human object(s). Furthermore, each detected point (x_t^d, y_t^d, z_t^d) is associated with a particular SNR for specifying the corresponding reception quality. To select reliable radar cloud points, we set the SNR threshold of 30 dB to filter out those unreliable points. However, the number of detected points D_t may sometimes be not sufficient to extract robust input features for indoor occupancy counting. Thus, we propose to “stack” M consecutive frames of point clouds to gather a large number of detected points in aggregate. The aforementioned stacked cloud points can be expressed by

$$C_i \stackrel{\text{def}}{=} \left\{ \left(x_{t'}^d, y_{t'}^d, z_{t'}^d \right) \middle| d = 0, 1, \dots, D_{t'} - 1, \right. \\ \left. t' = (i-1)N + 1, (i-1)N + 2, \dots, (i-1)N + M \right\} \quad (4)$$

where i specifies the stack index, and $i = 1, 2, \dots, I$. First, we collect M consecutive frames to obtain a stack of cloud points. Then we advance N frames to collect the next stack of cloud points and so on. Note that since $N \leq M$, a stack of cloud points and its next stack of cloud points may have overlapped frames. Then, we use such a sliding window to collect every I consecutive stacks of cloud points to form a “feature matrix” (to be further manifested later on). An example of the aforementioned stacking and sliding procedure for $M = 3$ and $N = 2$ is illustrated by Fig. 3.

B. Data Processing

In this section, we propose a novel robust feature extraction technique to obtain reliable input features, which are sophisticatedly distilled from the measured 3-D coordinates of the detected cloud points for indoor occupancy counting. First, a “voxelization” approach is applied to convert the measured 3-D coordinates of the detected cloud points into the grid-voxel units according to the “enhanced voxelization” scheme proposed in Section II-C of [40]. Then, a “peak

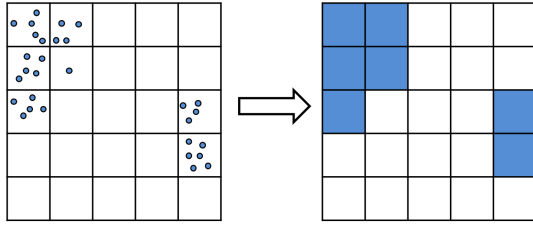


Fig. 4. Visualization example of the voxelization approach for a stacked point cloud C_j in the xy plane.

detection algorithm” is proposed to determine the number of reliable grid-voxel units in a stacked point cloud. Finally, robust features can be obtained by our proposed new feature extraction strategy. Our proposed data processing mechanism is detailed below.

1) **Voxelization:** According to [40], for each stacked point cloud C_i , the ranges of x -, y -, and z -axes can be determined as

$$K_i^x \stackrel{\text{def}}{=} \max\{x_{t'}^d\} - \min\{x_{t'}^d\} \quad (5)$$

$$K_i^y \stackrel{\text{def}}{=} \max\{y_{t'}^d\} - \min\{y_{t'}^d\} \quad (6)$$

$$K_i^z \stackrel{\text{def}}{=} \max\{z_{t'}^d\} - \min\{z_{t'}^d\} \quad (7)$$

where $d = 0, 1, \dots, D_t' - 1$ and $t' = (i-1)N + 1, (i-1)N + 2, \dots, (i-1)N + M$. Thus, for each stacked point cloud C_i , the resolutions of each grid-voxel unit along the x -, y -, and z -axes are given by

$$x_i^{\text{res}} \stackrel{\text{def}}{=} \frac{K_i^x}{x_{\text{cut}}} \quad (8)$$

$$y_i^{\text{res}} \stackrel{\text{def}}{=} \frac{K_i^y}{y_{\text{cut}}} \quad (9)$$

$$z_i^{\text{res}} \stackrel{\text{def}}{=} \frac{K_i^z}{z_{\text{cut}}} \quad (10)$$

where x_{cut} , y_{cut} , and z_{cut} specify the numbers of voxel-grid units along the x -, y -, and z -axes. In this work, we choose $x_{\text{cut}} = 6$ cm, $y_{\text{cut}} = 6$ cm, and $z_{\text{cut}} = 9$ cm according to our heuristic experience. For each stacked point cloud C_i , the values of x_i^{res} , y_i^{res} , and z_i^{res} represent the length, width, and height of each grid-voxel unit, respectively. A visualization example of the voxelization approach for a stacked point cloud C_i in the xy plane is illustrated by Fig. 4.

For each stacked point-cloud C_i , we can find s_i grid-voxel units, each of which contains at least a detected cloud point. In practice, one may collect the number of grid-voxel units, s_i , of each stacked point cloud C_i for I stacks of point clouds as given by

$$\mathcal{S} \stackrel{\text{def}}{=} \{s_i | i = 1, 2, \dots, I\}. \quad (11)$$

2) **Peak Detection:** To extract robust features to enhance the indoor occupancy counting performance, we propose a new multiple peak detection algorithm to acquire the local maxima of the time-series $\mathcal{S}$ as given by (11). The details of our proposed novel multiple peak detection algorithm for feature extraction are manifested by Algorithm 1. According

Algorithm 1 Multiple-Peak-Detection Algorithm for Feature Extraction

```

1 Input: the time-series  $\mathcal{S}$ ;
2 Output: the index set  $\mathcal{P}$  of local maxima of the input time-series  $\mathcal{S}$ ;
3 Function  $\mathcal{P} = \text{multiple-peak-detection}(\mathcal{S})$ 
4   Initialization:
5    $\mathcal{P} = \emptyset$ ;
6   the temporary local maximum of the input time-series  $\mathcal{S}$  is set to be  $s_{\text{tmp}} = 0$ ;
7   the sample index of the temporary local maximum of the input time-series  $\mathcal{S}$  is set to be  $p_{\text{tmp}} = 1$ ;
8   The threshold is set to be  $\gamma = 3$  for multiple peak detection;
9   For  $i = 1$  to  $I$ , do the following:
10    if  $s_{i+1} > s_i$  then
11       $s_{\text{tmp}} = s_{\text{tmp}} + 1$ ;
12    else
13      if  $s_{\text{tmp}} > \gamma$  then
14         $p_{\text{tmp}} = i$ ;
15         $s_{\text{tmp}} = 0$ ;
16      else
17        if  $s_i > s_{p_{\text{tmp}}}$  then
18           $p_{\text{tmp}} = i$ ;
19        end
20      end
21      if  $p_{\text{tmp}} \neq 1$  and  $p_{\text{tmp}} \notin \mathcal{P}$  then
22         $\mathcal{P} = \mathcal{P} \cup \{p_{\text{tmp}}\}$ ;
23      end
24    end
25  return  $\mathcal{P}$ .
26 End Function

```

to Algorithm 1, the index set $\mathcal{P}$ of the local maxima of the time-series $\mathcal{S}$ can be obtained as

$$\mathcal{P} \stackrel{\text{def}}{=} \{p_j | j = 1, 2, \dots, J\} \quad (12)$$

where J denotes the number of local maxima one can obtain from the original time-series $\mathcal{S}$. Then, the average ϕ_j of all the sample values in the original time-series $\mathcal{S}$ between two adjacent local maxima s_{p_j} and $s_{p_{j+1}}$ can be calculated as

$$\phi_j \stackrel{\text{def}}{=} \frac{s_{p_j} + s_{p_{j+1}} + \dots + s_{p_{j+1}}}{p_{j+1} - p_j + 1} \quad (13)$$

where p_j denotes the sample index of the original time-series $\mathcal{S}$ corresponding to the j th local maximum.

Nineteen collected stacks of cloud points are illustrated as an example to show how our proposed multiple peak detection algorithm works by Fig. 5. The detected local maxima of the 19 collected stacks of cloud points found by Algorithm 1 are highlighted as red circles in Fig. 5. The corresponding sample index set $\mathcal{P}$ of the local maxima in Fig. 5 is $\mathcal{P} = \{2, 5, 10, 15, 19\}$. Furthermore, the average ϕ_j of all the sample values between two adjacent local maxima can be calculated according to (13). Thus, the averages between each

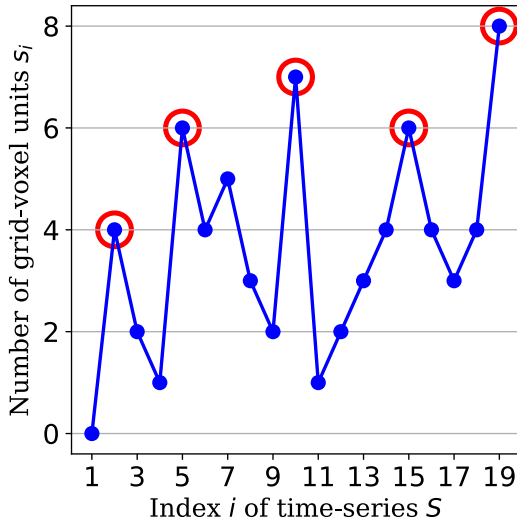


Fig. 5. Number of grid-voxel units s_i resulting from the 19 stacked point clouds with respect to the sample index i .

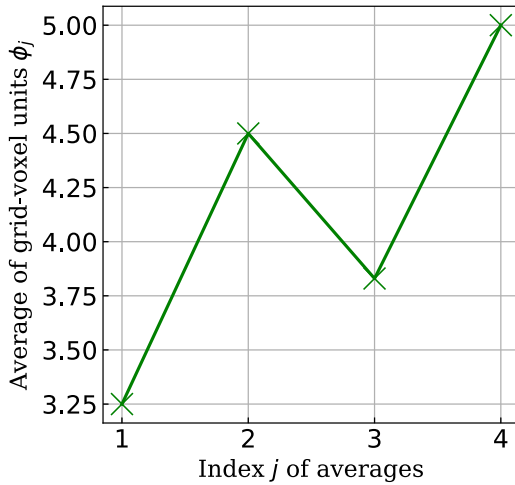


Fig. 6. Averages ϕ_j for the example shown by Fig. 5 with respect to the local maximum index j .

pair of adjacent local maxima in Fig. 5 can be determined and shown in Fig. 6.

3) **Feature Extraction:** For each pair of adjacent average values ϕ_j and ϕ_{j+1} obtained according to (13), we calculate two additional features u_j and v_j such that

$$u_j \stackrel{\text{def}}{=} \phi_{j+1} - \phi_j \quad (14)$$

$$v_j \stackrel{\text{def}}{=} \frac{1}{\phi_{j+1} - \phi_j}. \quad (15)$$

Then, the j th feature vector $\vec{\omega}_j$ can be formed as

$$\vec{\omega}_j \stackrel{\text{def}}{=} [\phi_j \ \phi_{j+1} \ u_j \ v_j]^\top. \quad (16)$$

Now we may establish the training feature matrix, say Ω_{tr} , from the training dataset and the test feature matrix, say Ω_{te} , from the test dataset such that

$$\Omega_{\text{tr}} \stackrel{\text{def}}{=} [\vec{\omega}_{\bar{j}}, \vec{\omega}_{\bar{j}+1}, \dots, \vec{\omega}_{\bar{j}+J_{\text{tr}}-2}] \quad (17)$$

and

$$\Omega_{\text{te}} \stackrel{\text{def}}{=} [\vec{\omega}_{\check{j}}, \vec{\omega}_{\check{j}+1}, \dots, \vec{\omega}_{\check{j}+J_{\text{te}}-2}] \quad (18)$$

where $\bar{j}$ and $\check{j}$ denote the starting feature vector indices of the training and test data, respectively; $J_{\text{tr}} - 1$ and $J_{\text{te}} - 1$ specify the total numbers of feature vectors collected from the training and test datasets, respectively.

C. KNN Classifier

In this work, we formulate the on-line indoor occupancy counting problem as a “multiclassification” problem for H classes (Class h implies that there are h occupants for $h = 0, 1, \dots, H-1$). Then, a commonly adopted effective machine learning tool, namely, KNN classifier (see [41]), is invoked in this work. The aforementioned feature matrices Ω_{tr} for training and Ω_{te} for testing are the input data to the KNN classifier. The KNN classifier will find the k nearest (most similar) training data (feature vectors) to a particular test feature vector so as to estimate the total number of occupants by the majority vote among these KNNs. In this work, k is chosen to be 30 according to our heuristic experience.

IV. EXPERIMENTS

In this work, we focus on the data acquired by the TI-IWR6843 mmWave FMCW radar. The floor plan, layout, and room dimensions comply with the previous descriptions presented in Section III-A. Four hypotheses (i.e., $H = 4$) are stipulated in this work, namely, $h = 0$, $h = 1$, $h = 2$, and $h = 3$, which represent “no occupant,” “one occupant,” “two occupants,” and “three occupants,” respectively. To mimic the real-world scenario such as a nursing assistant looking after one or more patients, we design three complex test scenarios as follows.

- 1) No occupant (neither nurse nor patient) is in the activity range (within the room) for 1 min and then a person steps in the activity range and then walks around for 1 min (see Fig. 7(a) and (b) for illustration).
- 2) First, two occupants walk around and they keep distance from each other in the meantime for 1 min. Then, one of them keeps walking around while the other stands still remotely for 1 min. Finally, both of them walk apart from each other for 1 min and then they walk side by side for 1 min (see Fig. 7(c) and (d) for illustration).
- 3) First, a person walks around in the activity range for 1 min. Then, two people walk apart from each other in the activity range for 1 min. Finally, three persons walk apart from each other in the activity range for 1 min (see Fig. 7(e) for illustration).

The aforementioned three scenarios are performed in sequence for 9 min totally.

To justify the robustness of our proposed new scheme, we also include some extra furniture as *obstacles* in the experiment area to make the test scenarios much more realistic. The three new test scenarios are the same as previously described with the only difference in that we add one desk and two chairs around the upper right corner of the experiment area (aggregately denoted by the red circle in Fig. 8) and one more



Fig. 7. Test scenarios designed in our experiment without any obstacle. (a) No occupant is in the activity range. (b) Occupant is walking in the activity range. (c) Two occupants are walking apart from each other in the activity range. (d) Two occupants are walking side by side in the activity range. (e) Three occupants are walking apart from each other in the activity range. (f) Three occupants are walking side by side in the activity range.

chair around the lower left corner (denoted by the blue circle in Fig. 8) as obstacles. The three test scenarios consisting of the aforementioned furniture as obstacles are shown by Fig. 8.

To produce the diversity in our experiments, the mmWave radar data are collected from 12 different people (ten men and two women). We randomly assign these 12 persons into four groups, each of which contains three people. Each group performs the above-stated three scenarios once. The frame rate of the mmWave radar is 10 frames/s. As each sequence lasts for 9 min, we collect 5400 frames from each group. The collected mmWave radar data contain 21 600 frames from four groups totally. Then we split these 21 600 frames chronologically into the training data and the test data subject to the division ratio of 8:2. Thus, the training data contain 17 280 frames while the test data contain 4320 frames. After stacking $M = 3$ frames and sliding $N = 2$ frames each time over the consecutive training and test data frames as stated in Section III-A, we have 10 730 stacks of cloud points in the training dataset and 2752 stacks of cloud points in the test dataset. We also study the effect of different values of M and N on the performance of our proposed indoor occupancy counting approach in Section IV-C later on. After carrying out the data processing step as stated in Section III-B, we obtain a 1444×4 training feature matrix Ω_{tr} and an 870×4 test feature matrix Ω_{te} subject to $J_{tr} = 1445$ and $J_{te} = 871$.

A. Performance Comparison

Here, we compare two other existing indoor occupancy counting schemes based on the clustering techniques, namely, the “WIMOC” and “cTracker” (see [31], [32]), and

the conventional indoor occupancy counting method called “Gtrack” provided by TI with our proposed new indoor occupancy counting approach. For notational simplicity, our proposed new scheme is denoted by “Voxelization-KNN.” For a fair quantitative evaluation, we use the same training dataset to train the aforementioned four indoor occupancy counting methods and take the accuracy (ACC) measure to evaluate the individual performances over the test dataset. According to [42], the definition of ACC is given by

$$\text{ACC} \stackrel{\text{def}}{=} \frac{1}{H} \sum_{h=0}^{H-1} \frac{\text{TP}_h + \text{TN}_h}{\text{TP}_h + \text{FN}_h + \text{FP}_h + \text{TN}_h} \quad (19)$$

where TP_h specifies the total number of true positives when there are actually h occupants, TN_h denotes the total number of true negatives when there are actually h occupants, FP_h represents the total number of false positives when there are actually h occupants, and FN_h denotes the total number of false negatives when there are actually h occupants.

The performances of our proposed new “Voxelization-KNN” scheme and the existing “WIMOC,” “cTracker,” and “Gtrack” schemes in terms of ACC over the test datasets corresponding to the scenarios without any obstacle in the experiment area and those corresponding to the scenarios consisting of some furniture as obstacles in the experiment area are listed by Table II. According to Table II, our proposed new “Voxelization-KNN” indoor occupancy counting method always leads to the highest accuracies over the test dataset corresponding to the scenarios without any obstacle in the experiment area. On the other hand, for the scenarios consisting of some furniture as obstacles in the experiment

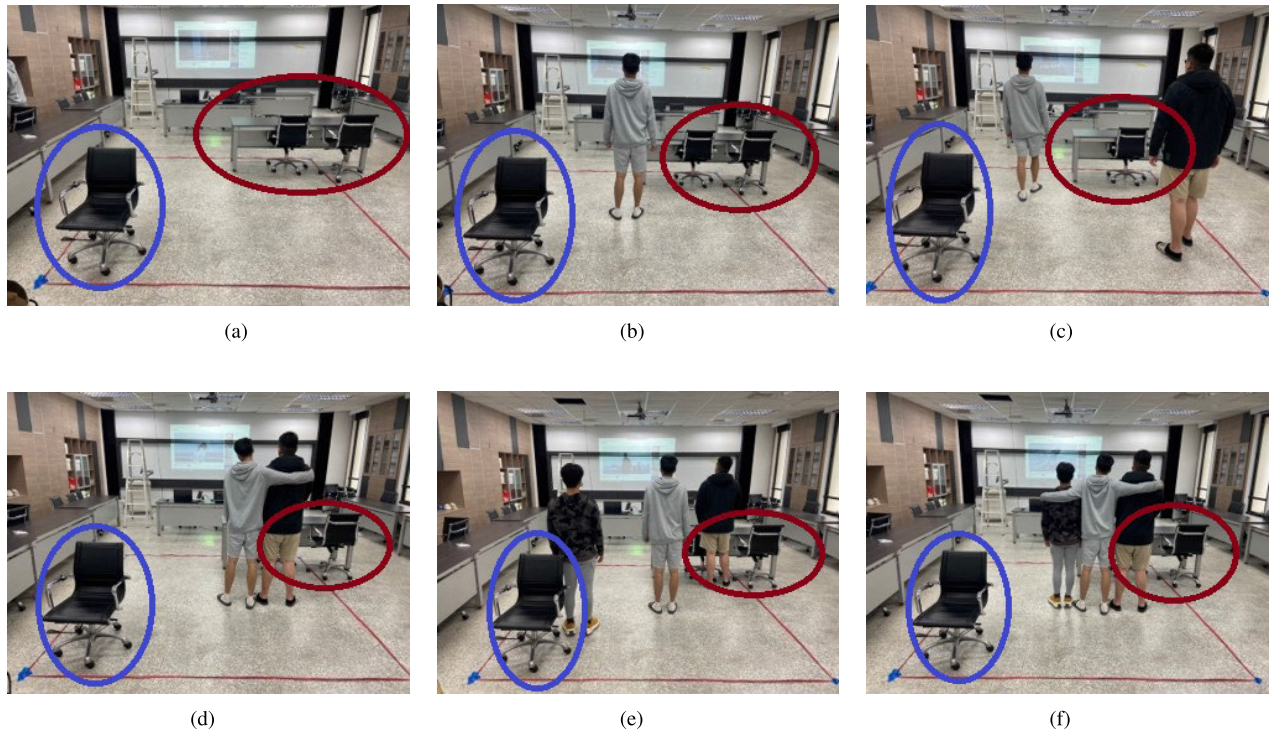


Fig. 8. Test scenarios designed in our experiment consisting of some furniture as obstacles. (a) No occupant is in the activity range. (b) Occupant is walking in the activity range. (c) Two occupants are walking apart from each other in the activity range. (d) Two occupants are walking side by side in the activity range. (e) Three occupants are walking apart from each other in the activity range. (f) Three occupants are walking side by side in the activity range.

area, the accuracies of our proposed new “Voxelization-KNN” indoor occupancy counting method slightly degrade due to the complexity of the experiment environment. However, our proposed new “Voxelization-KNN” scheme still leads the highest accuracies for $h = 0$, $h = 2$, and $h = 3$ occupants. The accuracies of the “WIMOC” scheme for the test scenarios consisting of obstacles in the experiment area slightly improve for $h = 1$ occupant but significantly degrade for $h = 2$ and $h = 3$ occupants. Similarly, the accuracies of the “cTracker” and “Gtrack” schemes improve for $h = 2$ occupant but significantly degrade for $h = 3$ occupants. Since the existing “WIMOC,” “cTracker,” and “Gtrack” schemes are not robust against the change(s) in the experiment environment. According to Table II, our proposed new “Voxelization-KNN” method greatly outperforms the existing “WIMOC,” “cTracker,” and “Gtrack” schemes for most test scenarios.

Finally, the confusion matrices resulting from our proposed new “Voxelization-KNN” approach for indoor occupancy counting for the test scenarios without any obstacle and for the test scenarios consisting of some obstacles in the experiment area are presented by Fig. 9. The confusion matrix indicates the matching probabilities between the estimated numbers of occupants by our “Voxelization-KNN” classifier and the actual numbers of occupants. A confusion matrix, say B , is given by

$$B \stackrel{\text{def}}{=} \begin{bmatrix} b_{0,0} & b_{0,1} & \cdots & b_{0,H-1} \\ b_{1,0} & b_{1,1} & \cdots & b_{1,H-1} \\ \vdots & \vdots & \vdots & \vdots \\ b_{H-1,0} & b_{H-1,1} & \cdots & b_{H-1,H-1} \end{bmatrix} \quad (20)$$

TABLE II

ACCURACIES RESULTING FROM OUR PROPOSED NEW “VOXELIZATION-KNN” METHOD AND THE EXISTING “WIMOC,” “CTRACKER,” AND “GTRACK” SCHEMES IN THE PRESENCE OF DIFFERENT NUMBERS OF OCCUPANTS h FOR TWO DIFFERENT TEST ENVIRONMENTS

	h	cTracker (%)	WIMOC (%)	Gtrack (%)	Voxelization-KNN (%)
No obstacle in the experiment area	0	99.0	98.2	97.2	100
	1	91.3	89.2	88.6	97.8
	2	46.0	42.0	29.8	97.2
	3	69.4	45.8	37.2	88.5
	Average	76.4	68.8	63.2	95.8
	h	cTracker (%)	WIMOC (%)	Gtrack (%)	Voxelization-KNN (%)
Some obstacles in the experiment area	0	100.0	100.0	100.0	100.0
	1	83.6	99.0	96.0	94.5
	2	65.0	35.0	59.0	92.1
	3	2.0	26.0	32.0	68.1
	Average	62.7	65.0	71.8	88.7

where $b_{h,h'}$ denotes the percentage of test samples from which our “Voxelization-KNN” scheme estimates that there are h' occupants while there are actually h occupants. Hence, the diagonal entries $b_{h,h}$ of B , $h = 0, 1, \dots, H - 1$ correspond to the probabilities of correct estimation. According to Fig. 9, our proposed new “Voxelization-KNN” approach leads to almost perfect performance for zero, one, and two occupants for different experiment environments. Our proposed new “Voxelization-KNN” approach also leads to very good performance for the scenarios consisting of three occupants in different experimental environments.

For further quantitative evaluation, we analyze the performances of our proposed new “Voxelization-KNN” approach and three other existing methods, i.e., the “WIMOC,” “cTracker,” and “Gtrack” schemes for four complex test

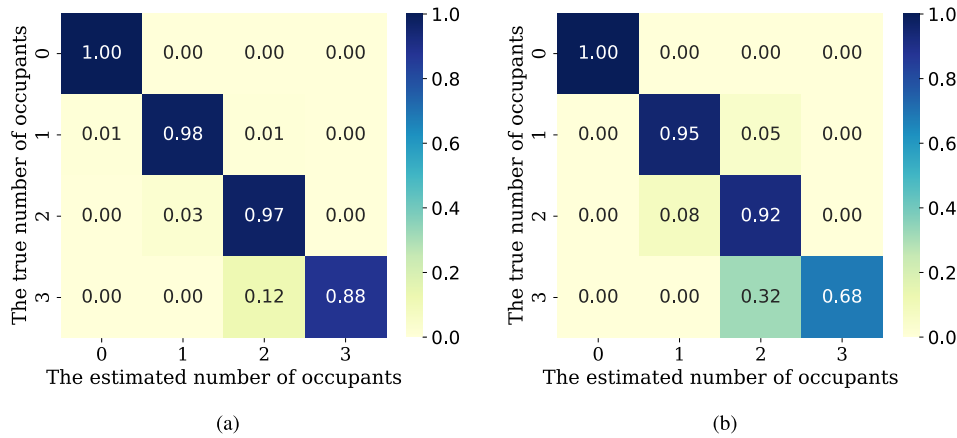


Fig. 9. Confusion matrices resulting from our proposed new “Voxelization-KNN” approach for indoor occupancy counting for different experimental environments. (a) Confusion matrix for the test scenarios without any obstacle in the experiment area. (b) Confusion matrix for the test scenarios consisting of some furniture as obstacles in the experiment area.

scenarios involving two or three occupants. These four test scenarios are designed to mimic real-world situations and detailed as follows.

- 1) Two occupants walk around and they keep distance from each other in the meantime for 1 min [illustrated by Fig. 7(c)].
- 2) Two occupants walk side by side within the activity range for 1 min [illustrated by Fig. 7(d)].
- 3) Three people walk apart from each other in the activity range for 1 min [illustrated by Fig. 7(e)].
- 4) Three people walk side by side in the activity range for 1 min [illustrated by Fig. 7(f)].

Similarly, we also create four new test scenarios by adding some furniture as the obstacles in the experiment area to evaluate the robustness of our proposed new method. The four new test scenarios are the same as aforementioned with the only difference in that we add one desk and two chairs around the upper right corner in the experiment area and one more chair around the lower left corner as obstacles as shown by Fig. 8. For notational simplicity, the above-stated four test scenarios are denoted by “two apart,” “two close,” “three apart,” and “three close,” respectively. The frame rate of the mmWave radar is 10 frames/s and a group performs such four test scenarios for 4 min totally. Thus, we can collect 2400 frames from these new test scenarios. A pretrained KNN classifier using the aforementioned training feature matrix Ω_{tr} is used to test these new scenarios.

The performances of our proposed new “Voxelization-KNN” approach along with the existing “WIMOC,” “cTracker,” and “Gtrack” schemes in terms of ACC for the aforementioned four test scenarios in two different experimental environments are listed by Table III. According to Table III, our proposed new “Voxelization-KNN” method always leads to the highest accuracies among all for each of the aforementioned four test scenarios both without any obstacle in the experiment area and consisting of some furniture as obstacles in the experiment area. Nonetheless, our proposed new method still registers a low accuracy of 0.363 in the absence of obstacle and a low accuracy of 0.546 in the

TABLE III
ACCURACIES RESULTING FROM OUR PROPOSED NEW “VOXELIZATION-KNN” METHOD ALONG WITH THE EXISTING “WIMOC,” “CTRACKER,” AND “GTRACK” SCHEMES FOR TWO DIFFERENT EXPERIMENTAL ENVIRONMENTS OVER FOUR COMPLEX TEST SCENARIOS

Scenarios	cTracker (%)	WIMOC (%)	Gtrack (%)	Voxelization-KNN (%)
No obstacle in the experiment area	Two apart	76.2	74.9	39.6
	Two close	7.1	4.6	20
	Three apart	59.8	39.6	18.6
	Three close	0.0	0.0	9.3
	Average	35.8	29.8	21.9
				75.8
Some obstacles in the experiment area	Two apart	43.0	64.0	65.0
	Two close	3.3	5.0	10.0
	Three apart	8.0	15.0	29.0
	Three close	1.0	14.0	21.0
	Average	13.8	24.5	31.3
				79.2

presence of some furniture as obstacles, respectively, for the “three close” scenario. The accuracy for the “three close” scenario in the presence of some obstacles is higher than the accuracy for the “three close” scenario in the absence of obstacle because some obstacles might be misrecognized as occupants occasionally. Furthermore, the three existing “WIMOC,” “cTracker,” and “Gtrack” schemes also fail for the “three close” scenario in both the experimental environments (with and without obstacles). Consequently, our proposed new “Voxelization-KNN” indoor occupancy counting method greatly outperforms the existing “WIMOC,” “cTracker,” and “Gtrack” schemes for the four complex test scenarios in both the experimental environments (with and without obstacles).

To understand the difficulty arising from the “three close” scenario, let us visualize the corresponding point clouds. First, a stacked point cloud C_i extracted from the “two close” scenario by our proposed new “Voxelization-KNN” scheme can be visualized as Fig. 10(a). The point clouds for the “two close” scenario resulting from the existing “WIMOC” and “cTracker” schemes can be visualized as Fig. 10(b) and (c), respectively. According to Fig. 10(a)–(c), the stacked point cloud resulting from the our proposed new “Voxelization-KNN” approach can be well split into two separate clusters while the point clouds resulting from the existing “WIMOC” and “cTracker” schemes tend to be more difficult to be

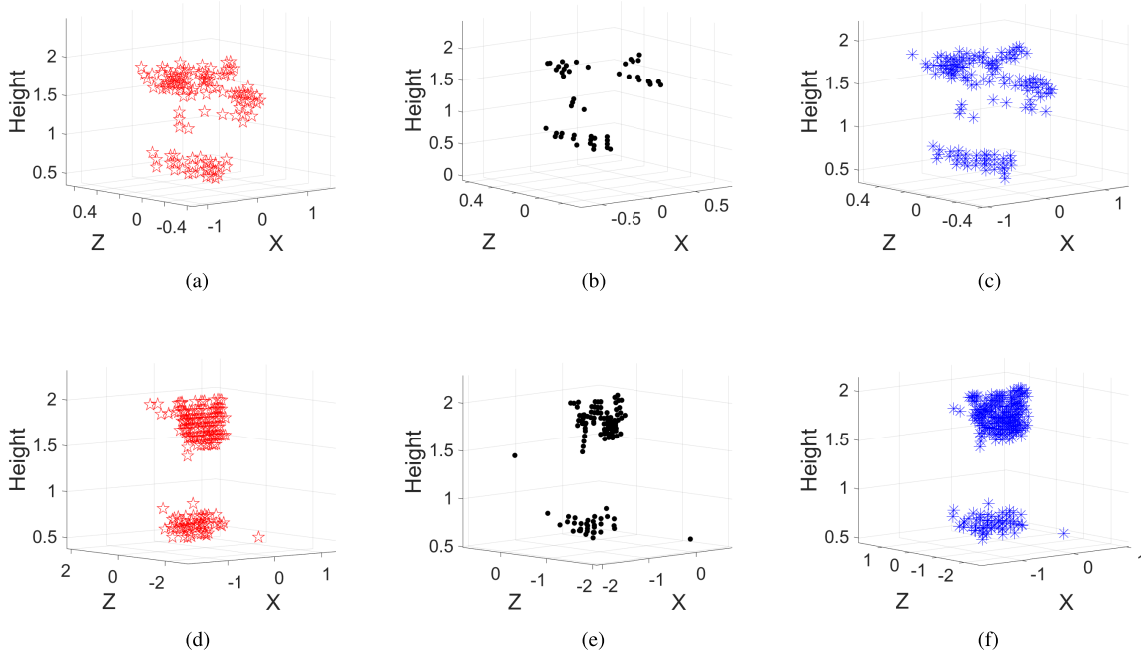


Fig. 10. Stacked point clouds produced by our proposed new “Voxelization-KNN” method along with the existing “WIMOC” and “cTracker” schemes for the “two close” and “three close” scenarios. (a) Our proposed new “Voxelization-KNN” method for the “two close” scenario. (b) Existing cTracker scheme for the “two close” scenario. (c) Existing WIMOC scheme for the “two close” scenario. (d) Our proposed new “Voxelization-KNN” method for the “three close” scenario. (e) Existing cTracker scheme for the “three close” scenario. (f) Existing WIMOC scheme for the “three close” scenario.

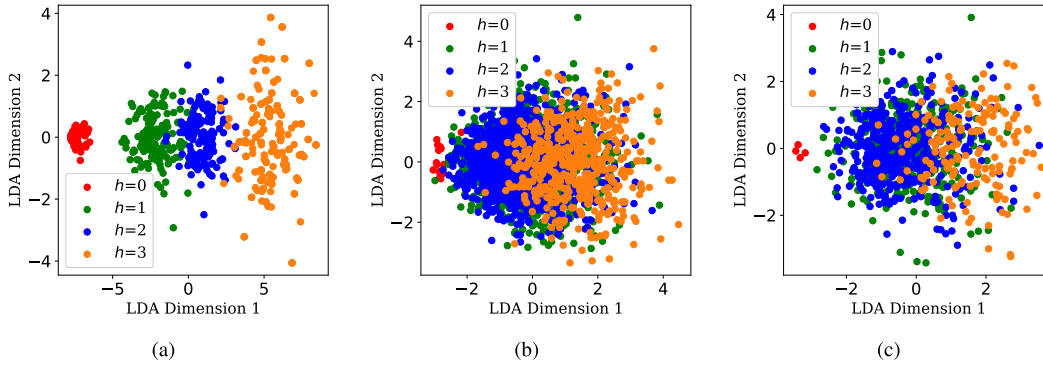


Fig. 11. Visualization for the LDA over various training features resulting from our proposed new Voxelization-KNN approach along with the existing cTracker and WIMOC schemes. (a) Visualization of the training features resulting from our proposed new Voxelization-KNN scheme using LDA. (b) Visualization of the training features resulting from the existing cTracker scheme using LDA. (c) Visualization of the training features resulting from the existing WIMOC scheme using LDA.

well-clustered. As a result, our proposed new “Voxelization-KNN” scheme outperforms the existing “WIMOC” and “cTracker” schemes for the “two close” test scenario. Besides, we can also visualize a stacked point cloud C_i for the “three close” scenario resulting from our proposed new “Voxelization-KNN” approach as Fig. 10(d). The point clouds for the “three close” scenario resulting from the existing “WIMOC” and “cTracker” schemes can be visualized as Fig. 10(e) and (f), respectively. According to Fig. 10(d)–(f), none of the stacked point cloud produced by our proposed new “Voxelization-KNN” method and the point clouds produced by the existing “WIMOC” and “cTracker” schemes can be separated into three clusters easily. Consequently, the accuracies for the “three close” scenario resulting from the three aforementioned schemes are all very low.

B. Robustness Analysis for Features in Comparison

Linear discriminant analysis (LDA) is a common statistical tool for us to explore the “linear separability of multiple classes” in a particular feature space [43]. For the training feature matrix Ω_{tr} obtained by (17), we have $J_{tr}-1$ feature vectors (samples) $\vec{\omega}_j$ given by (16). The training feature matrix Ω_{tr} consists of H classes ($H = 4$ here). A unit vector $\vec{\eta}$ ($\|\vec{\eta}\| = 1$ where $\|\cdot\|$ denotes the vector norm) should be determined to represent the *optimal* projection direction that “best” distinguishes the H underlying classes. According to [43], such a unit vector $\vec{\eta}$ needs to satisfy the following equation:

$$\Upsilon_w^{-1} \Upsilon_b \vec{\eta} = \lambda \vec{\eta} \quad (21)$$

where Υ_w^{-1} is the inverse of the within-class scatter matrix Υ_w , and Υ_b is the between-class scatter matrix. The within-class

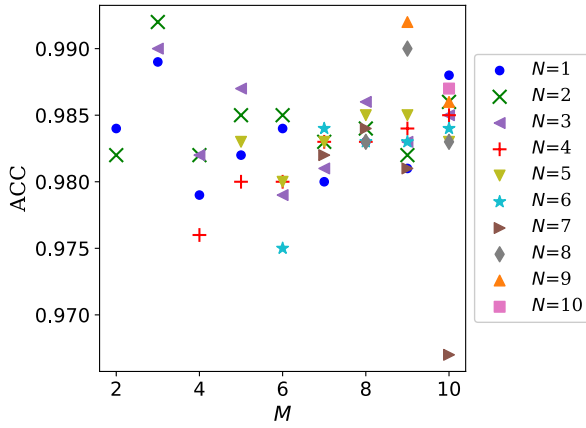


Fig. 12. Accuracies for indoor occupancy counting resulting from our proposed new “Voxelization-KNN” scheme subject to various M and N values.

scatter matrix Υ_w is defined by

$$\Upsilon_w \stackrel{\text{def}}{=} \sum_{h=0}^{H-1} \sum_{\tilde{\omega}_j \in h} (\tilde{\omega}_j - \tilde{\mu}_h) (\tilde{\omega}_j - \tilde{\mu}_h)^\top \quad (22)$$

where $\tilde{\mu}_h$ represents the mean vector of all the training feature vectors in Class h such that

$$\tilde{\mu}_h \stackrel{\text{def}}{=} \frac{1}{G_h} \sum_{\tilde{\omega}_j \in h} \tilde{\omega}_j \quad (23)$$

where G_h denotes the total number of training feature-vectors in Class h . Besides, the between-class scatter matrix Υ_b is defined by

$$\Upsilon_b \stackrel{\text{def}}{=} \sum_{h=0}^{H-1} G_h (\tilde{\mu}_h - \tilde{\mu}) (\tilde{\mu}_h - \tilde{\mu})^\top \quad (24)$$

where $\tilde{\mu}$ represents the mean vector of all the training feature vectors such that

$$\tilde{\mu} \stackrel{\text{def}}{=} \frac{1}{J_{\text{tr}} - 1} \sum_{j=1}^{J_{\text{tr}}-1} \tilde{\omega}_j. \quad (25)$$

Consequently, the LDA can be used to visualize the training features resulting from our proposed new “Voxelization-KNN” approach and the existing “WIMOC” and “cTracker” schemes.¹ For visualization convenience, we project the feature vectors onto a 2-D plane as illustrated by Fig. 11. According to Fig. 11, the training features resulting from our proposed new “Voxelization-KNN” method can be easily separated into four classes while the training features resulting from the existing “WIMOC” and “cTracker” schemes are difficult to be classified.

C. Effects of M and N

Finally, we analyze the effects of the number of stacked frames M and the number of sliding frames N on the training

¹As the Gtrack scheme is the built-in function of the TI mmWave radar board, no intermediate feature information is available.

TABLE IV
EXECUTION TIMES AND ACCURACIES SUBJECT TO
($M = 3, N = 2$), ($M = 9$, AND $N = 9$)

(M, N)	Execution Time (sec)	Accuracy (%)
(3, 2)	1.59	99.2
(9, 9)	6.77	99.2

performance of our proposed new indoor occupancy counting approach. In our experiments, we set $M = 2, 3, \dots, 10$ and $N = 1, 2, \dots, M$. Thus, we compare the respective performances of our proposed new “Voxelization-KNN” scheme in terms of ACC defined by (19) for the training dataset. The results are depicted by Fig. 12. According to Fig. 12, our proposed new “Voxelization-KNN” scheme reaches the highest training accuracy when ($M = 3, N = 2$) and ($M = 9, N = 9$). Furthermore, the execution times for training the KNN classifier subject to ($M = 3, N = 2$) and ($M = 9, N = 9$) are listed by Table IV. According to Table IV, the accuracies resulting from ($M = 3, N = 2$) are the same as those resulting from ($M = 9, N = 9$). However, the execution time subject to ($M = 3, N = 2$) is much shorter than that subject to ($M = 9, N = 9$). As a result, we propose to select ($M = 3, N = 2$) for our proposed new “Voxelization-KNN” approach for on-line applications.

V. CONCLUSION

In this article, we propose a novel robust indoor occupancy counting technique based on the data acquired by the TI-IWR6843 mmWave FMCW radar. The acquired radar data can be represented as sparse 3-D radar point clouds. Besides, we propose a novel feature extraction strategy to obtain the robust features for training the KNNs’ classifier. We convert the occupancy counting problem into a multiclassification problem such that the aforementioned KNN classifier can be adopted as the occupant counter. We compare our proposed new approach with three other existing methods, namely, “WIMOC,” “cTracker,” and “Gtrack.” Our proposed new method can reach up to a 95.8% accuracy according to our real-world experiments. Our proposed novel indoor occupancy counting scheme is demonstrated to be very promising for a wide variety of future applications including smart healthcare, intelligent surveillance, smart building control, management, etc.

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